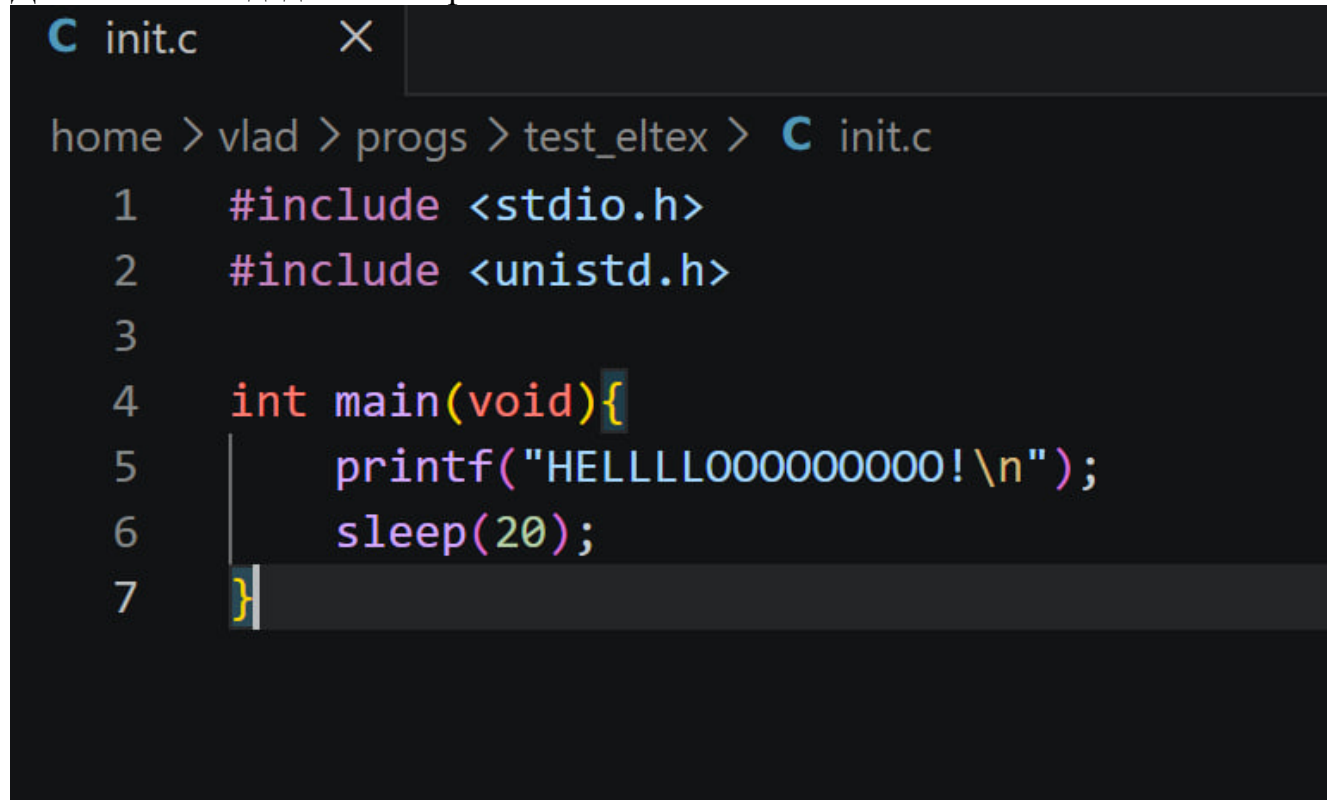


Создадим простенькую корневую файловую систему под нашей непростенькой wsl2.

Для начала создадим init.c файл

A screenshot of a code editor window titled "C init.c". The editor shows the following C code:

```
home > vlad > progs > test_eltex > C init.c
1  #include <stdio.h>
2  #include <unistd.h>
3
4  int main(void){
5      printf("HELLLL000000000!\n");
6      sleep(20);
7  }
```

И скомпилируем его под arm:

```
$ arm-linux-gnueabihf-gcc -static init.c -o init
```

< ~/progs/test_eltex .[^]>			
.n	Name	Size	Modify time
/..		UP--DIR	Jun 10 01:44
*init		424116	Jun 10 12:42
init.c		106	Jun 10 12:41
vexpress-v2p-ca9.dtb		14329	Jun 10 01:48
*zImage		11517K	Jun 10 01:42

Соберем нашу систему в архив:

```
$ echo init | cpio -o -H newc | gzip > initramfs.cpio.gz
```

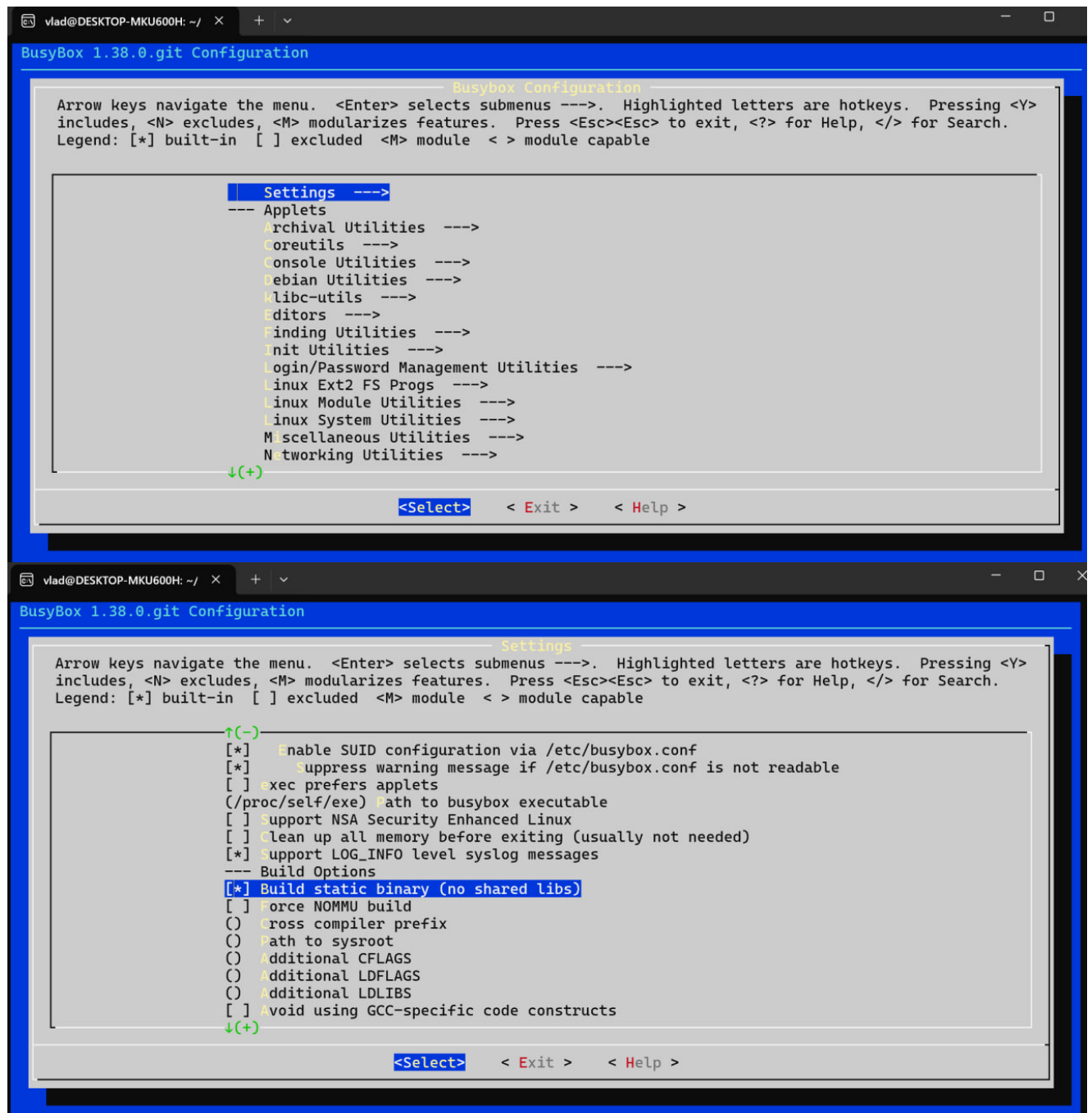
Запустим qemu:

```
$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -
kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-
ca9.dtb -append "console=ttyAMA0" -nographic
```

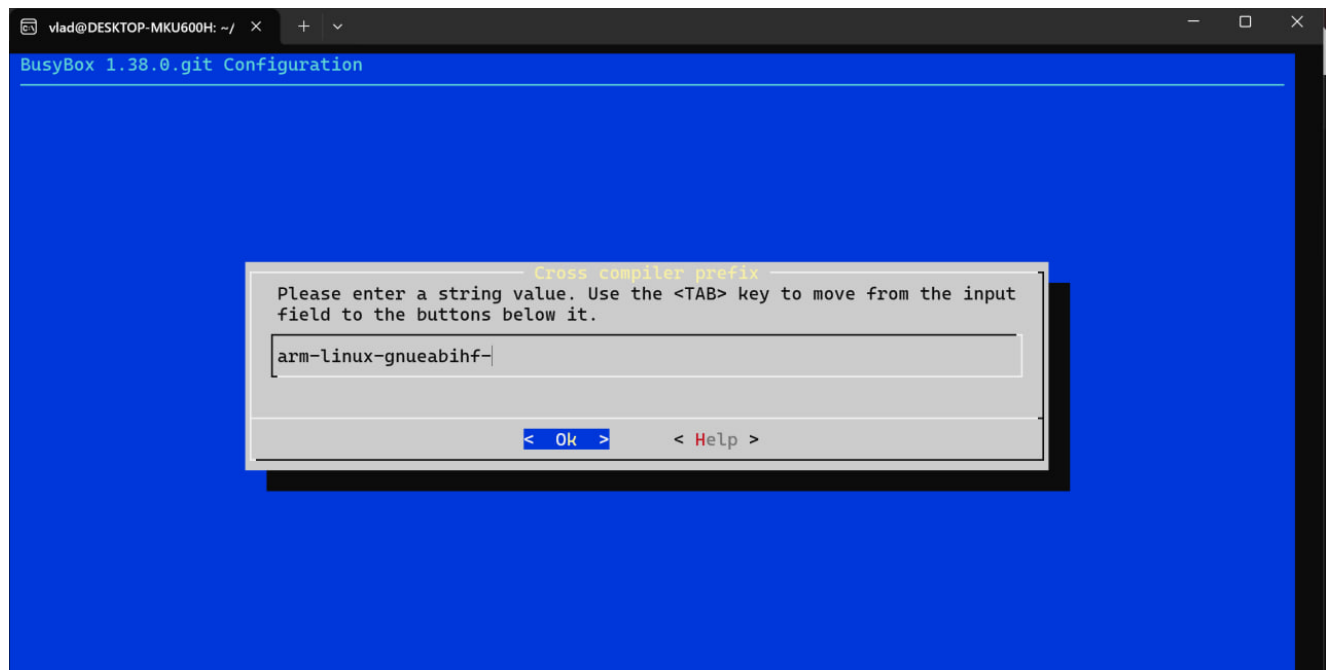
```

[ 1.719839] hw perfevents: enabled with armv7_cortex_a9 PMU driver, 7 (8000003f) counters available
[ 1.725503] NET: Registered PF_INET6 protocol family
[ 1.750014] Segment Routing with IPv6
[ 1.750334] In-situ OAM (IOAM) with IPv6
[ 1.750785] sit: IPv6, IPv4 and MPLS over IPv4 tunneling driver
[ 1.753963] NET: Registered PF_PACKET protocol family
[ 1.754011] can: controller area network core
[ 1.754331] NET: Registered PF_CAN protocol family
[ 1.754376] can: raw protocol
[ 1.754483] can: broadcast manager protocol
[ 1.754593] can: netlink gateway - max_hops=1
[ 1.756272] Key type dns_resolver registered
[ 1.756827] ThumbEE CPU extension supported.
[ 1.756889] Registering SWP/SWPB emulation handler
[ 1.810978] Loading compiled-in X.509 certificates
[ 1.830928] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[ 1.833856] printk: console [ttyAMA0] enabled
[ 1.865119] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
[ 1.914324] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
[ 1.916195] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[ 2.059456] input: ImEXPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/serio1/input/input1
[ 2.174159] input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10006000.kmi/serio0/input/input2
[ 3.186021] clk: Disabling unused clocks
[ 3.186328] PM: genpd: Disabling unused power domains
[ 3.235700] Freeing unused kernel image (initmem) memory: 2048K
[ 3.239629] Run /init as init process
HELLO000000000!

```



Пропишем компилятор для конфигурации, чтобы постоянно его не прописывать в команде:



Запускаем компиляцию:

\$ make -j\$(nproc)

```
vlad@DESKTOP-MKU600H:~/busybox$ make -j$(nproc)
SPLIT include/autoconf.h -> include/config/*
GEN include/bbconfigopts.h
GEN include/common_bufsiz.h
GEN include/embedded_scripts.h
HOSTCC applets/usage
HOSTCC applets/applet_tables
applets/usage.c: In function 'main':
applets/usage.c:52:17: warning: ignoring return value of 'write' declared with attribute 'warn_unused_result' [-Wunused-result]
   52 |             write(STDOUT_FILENO, usage_array[i].usage, strlen(usage_array[i].usage) + 1);
      |             ~~~~~^~~~~~
GEN include/usage_compressed.h
GEN include/applet_tables.h include/NUM_APPLETS.h
GEN include/applet_tables.h include/NUM_APPLETS.h
HOSTCC applets/usage_pod
CC applets/applets.o
LD applets/built-in.o
DOC busybox.pod
DOC BusyBox.txt
DOC busybox.1
```

Кажется, что все собралось, но я знаю, что нет.

```
CC libbb/xatouname.o
libbb/update_passwd.c: In function 'update_passwd':
libbb/update_passwd.c:153:17: warning: ignoring return value of 'fchown' declared with attribute 'warn_unused_result' [-Wunused-result]
   153 |             fchown(new_fd, sb.st_uid, sb.st_gid);
      |             ~~~~~^~~~~~
CC libbb/xconnect.o
CC libbb/xfunc_die.o
CC libbb/xfuncs.o
CC libbb/xfuncs_printf.o
CC libbb/xgetcwd.o
CC libbb/xgethostbyname.o
CC libbb/xreadlink.o
CC libbb/xrealloc_vector.o
CC libbb/xregcomp.o
AR libbb/lib.a
vlad@DESKTOP-MKU600H:~/busybox$
```

Нужно отключить модуль tc (Traffic Control), потому что с ним не работает.

Проверяем файл busybox. И видим, что он собрался под нашу архитектуру arm.

```
vlad@DESKTOP-MKU600H:~/busybox$ file busybox
busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1]=d0031e448c3bdeb4090103e824537f8c1c906165, for GNU/Linux 3.2.0, stripped
```

Продолжаем сборку:

\$ make install


```

./_install//usr/sbin/rdate -> ../../bin/busybox
./_install//usr/sbin/rdev -> ../../bin/busybox
./_install//usr/sbin/readahead -> ../../bin/busybox
./_install//usr/sbin/readprofile -> ../../bin/busybox
./_install//usr/sbin/remove-shell -> ../../bin/busybox
./_install//usr/sbin/rtcwake -> ../../bin/busybox
./_install//usr/sbin/seedrng -> ../../bin/busybox
./_install//usr/sbin/sendmail -> ../../bin/busybox
./_install//usr/sbin/setfont -> ../../bin/busybox
./_install//usr/sbin/setlogcons -> ../../bin/busybox
./_install//usr/sbin/svlogd -> ../../bin/busybox
./_install//usr/sbin/telnetd -> ../../bin/busybox
./_install//usr/sbin/tftpd -> ../../bin/busybox
./_install//usr/sbin/ubiattach -> ../../bin/busybox
./_install//usr/sbin/ubidetach -> ../../bin/busybox
./_install//usr/sbin/ubimkvol -> ../../bin/busybox
./_install//usr/sbin/ubirename -> ../../bin/busybox
./_install//usr/sbin/ubirmvol -> ../../bin/busybox
./_install//usr/sbin/ubirsvol -> ../../bin/busybox
./_install//usr/sbin/ubiupdatevol -> ../../bin/busybox
./_install//usr/sbin/udhcpd -> ../../bin/busybox

```

 You will probably need to make your busybox binary
 setuid root to ensure all configured applets will
 work properly.

vlad@DESKTOP-MKU600H:~/busybox\$ |

Собираем всю корневую файловую систему в архив:

\$ find . | cpio -o -H newc | gzip > initramfs.cpio.gz

```

vlad@DESKTOP-MKU600H:~/busybox/_install$ find . | cpio -o -H newc | gzip > initramfs.cpio.gz
3134 blocks
vlad@DESKTOP-MKU600H:~/busybox/_install$ |

```

Запускаем qemu:

**\$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -
 kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-
 ca9.dtb -append "console=ttyAMA0 rdinit=/bin/ash" -
 nographic**

```

2.562033] NET: Registered PF_PACKET protocol family
2.562080] can: controller area network core
2.562547] NET: Registered PF_CAN protocol family
2.562613] can: raw protocol
2.562757] can: broadcast manager protocol
2.562925] can: netlink gateway - max_hops=1
2.564534] Key type dns_resolver registered
2.565235] ThumbEE CPU extension supported.
2.565313] Registering SWP/SWPB emulation handler
2.634054] Loading compiled-in X.509 certificates
2.660007] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
2.663628] printk: console [ttyAMA0] enabled
2.728939] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
2.805421] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
2.808717] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
2.930799] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@400
00000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/serial/input/input2
4.603985] clk: Disabling unused clocks
4.604412] PM: genpd: Disabling unused power domains
4.669673] Freeing unused kernel image (initmem) memory: 2048K
4.675517] Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
#
# echo 1
#
# [ 15.218896] amba 1000f000.wdt: deferred probe pending: (reason unknown)
15.219069] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
15.219107] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
15.219132] amba 100e5000.watchdog: deferred probe pending: (reason unknown)

```

И все работает!

```

[15.219017] ksh: /bin/ash is init process
/bin/ash: can't access tty; job control turned off
~ #
~ # echo 1
1
~ # [ 15.218896] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 15.219069] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 15.219107] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 15.219132] amba 100e5000.watchdog: deferred probe pending: (reason unknown)

~ #
~ #
~ #
~ #
~ #
~ #
~ #
~ #
~ # ls
bin          initramfs.cpio.gz  root          usr
dev          linuxrc           sbin
~ #

```

Смонтируем proc и посмотрим запущенные процессы:

```
$ mkdir proc
```

```
$ mount -t proc proc proc
```

```

[15.219575] Freeing unused kernel image (initramfs) memory: 2610K
~ # ps
PID   USER     TIME   COMMAND
    1  0         0:01   /bin/ash
    2  0         0:00   [kthreadd]
    3  0         0:00   [pool_workqueue_]
    4  0         0:00   [kworker/R-rcu_g]
    5  0         0:00   [kworker/R-sync_]
    6  0         0:00   [kworker/R-kvfre]
    7  0         0:00   [kworker/R-slub_]
    8  0         0:00   [kworker/0:0-eve]
    9  0         0:00   [kworker/0:0H-ev]
   10  0         0:00   [kworker/0:1-rcu]
   11  0         0:02   [kworker/u16:0-e]
   12  0         0:00   [kworker/R-mm_pe]
   13  0         0:00   [ksoftirqd/0]
   14  0         0:00   [rcu_sched]
   15  0         0:00   [rcu_exp_par_gp_]

```